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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026

Name: Raya Llyanne Bernabe Palmes	SCORE:
Section: BSIS01	Date: 10/12/ 2026

1. O(1) — Constant Time CODE

```
using System;
namespace O_1__Constant_Time__Example_1_
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void label1_Click(object sender, EventArgs e)
        {

        }

        private void label1_Click_1(object sender, EventArgs e)
        {

        }

        private void lblDescription1_Click(object sender, EventArgs e)
        {

        }

        private void btnRun_Click(object sender, EventArgs e)
        {
            int[] numbers = { 10, 20, 30, 40, 50 };
            int answer = numbers[4];

            Console.WriteLine(answer);
            MessageBox.Show(answer.ToString());
        }

        private void txtIndex_TextChanged(object sender, EventArgs e)
        {

        }

        private void txtNumbers_TextChanged(object sender, EventArgs e)
        {

        }
```

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ANSWER SHEET
Term 1 AY 2025 - 2026

```
45.
46.         private void lbl2_Click(object sender, EventArgs e)
47.         {
48.
49.         }
50.
51.         private void button1_Click(object sender, EventArgs e)
52.         {
53.             int a = 10;
54.             int b = 20;
55.             int sum = a + b;
56.             Console.WriteLine(sum);
57.             MessageBox.Show(sum.ToString());
58.         }
59.
60.         private void panel1_Paint(object sender, PaintEventArgs e)
61.         {
62.
63.         }
64.
65.         private void panel3_Paint(object sender, PaintEventArgs e)
66.         {
67.
68.         }
69.
70.         private void button2_Click(object sender, EventArgs e)
71.         {
72.             string[] fruits = { "Strawberry", "Cranberry", "Blueberry" };
73.             string fruit = fruits[2];
74.
75.             Console.WriteLine(fruit);
76.             MessageBox.Show(fruit);
77.         }
78.
79.         private void label8_Click(object sender, EventArgs e)
80.         {
81.
82.         }
83.
84.         private void label15_Click(object sender, EventArgs e)
85.         {
86.
87.         }
88.
89.         private void label11_Click(object sender, EventArgs e)
90.         {
91.
92.         }
93.
94.         private void label21_Click(object sender, EventArgs e)
95.         {
96.
97.         }
```

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ANSWER SHEET
Term 1 AY 2025 - 2026

```
98.     }
99. }
```

O(1) Constant Time (Example 1)

```
21      private void lblDescription1_Click(object sender, EventArgs e)
22      {
23
24      }
25
        1 reference
26      private void btnRun_Click(object sender, EventArgs e)
27      {
28          int[] numbers = { 10, 20, 30, 40, 50 };
29          int answer = numbers[4];
30
31          Console.WriteLine(answer);
32          MessageBox.Show(answer.ToString());
33      }
34
        0 references
35      private void txtIndex_TextChanged(object sender, EventArgs e)
36      {
37
38      }
39
        0 references
40      private void txtNumbers_TextChanged(object sender, EventArgs e)
41      {
42
43      }
44
        1 reference
45      private void lbl2_Click(object sender, EventArgs e)
46      {
47
48      }
49
        1 reference
50      private void button1_Click(object sender, EventArgs e)
51      {
52          int a = 10;
53          int b = 20;
54          int sum = a + b;
55          Console.WriteLine(sum);
56          MessageBox.Show(sum.ToString());
57      }
58
        0 references
59      private void panel1_Paint(object sender, PaintEventArgs e)
60      {
61
62      }
63
        0 references
64      private void panel3_Paint(object sender, PaintEventArgs e)
65      {
66
67      }
68
        1 reference
69      private void button2_Click(object sender, EventArgs e)
70      {
71          string[] fruits = { "Strawberry", "Cranberry", "Blueberry" };
72          string fruit = fruits[2];
73
74          Console.WriteLine(fruit);
75          MessageBox.Show(fruit);
76      }
77
        0 references
78      private void label8_Click(object sender, EventArgs e)
79      {
80
```

47 % No issues found

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ANSWER SHEET
Term 1 AY 2025 - 2026

O(1) Constan...ple 1).csproj | Form1.cs* | Form1.cs [Design]*

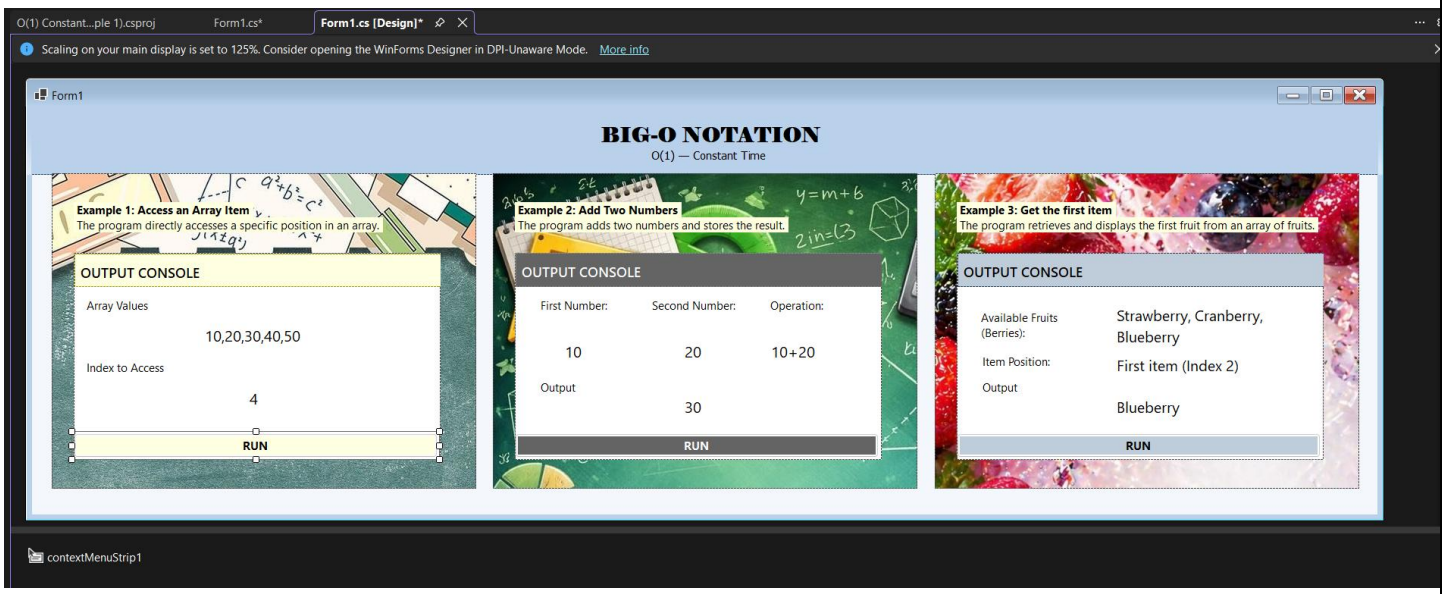
```
<Project Sdk="Microsoft.NET.Sdk">

  <PropertyGroup>
      <OutputType>Exe</OutputType>
      <TargetFramework>net10.0-windows</TargetFramework>
    <RootNamespace>O_1__Constant_Time__Example_1_</RootNamespace>
    <Nullable>enable</Nullable>
    <UseWindowsForms>true</UseWindowsForms>
    <ImplicitUsings>enable</ImplicitUsings>
  </PropertyGroup>

  <ItemGroup>
    <Compile Update="Properties\Resources.Designer.cs">
      <DesignTime>True</DesignTime>
      <AutoGen>True</AutoGen>
      <DependentUpon>Resources.resx</DependentUpon>
    </Compile>
  </ItemGroup>

  <ItemGroup>
    <EmbeddedResource Update="Properties\Resources.resx">
      <Generator>ResXFileCodeGenerator</Generator>
      <LastGenOutput>Resources.Designer.cs</LastGenOutput>
```

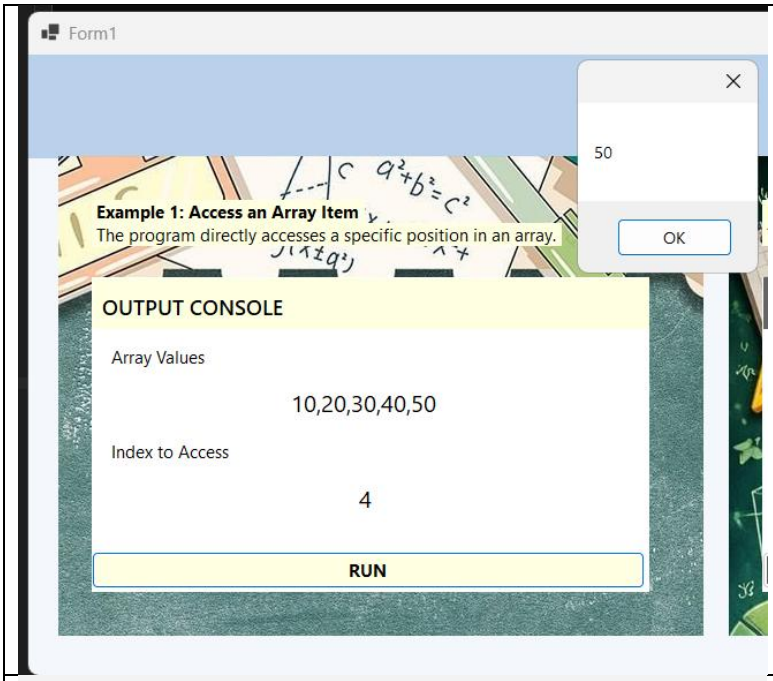
2. O(1) — Constant Time DESIGN

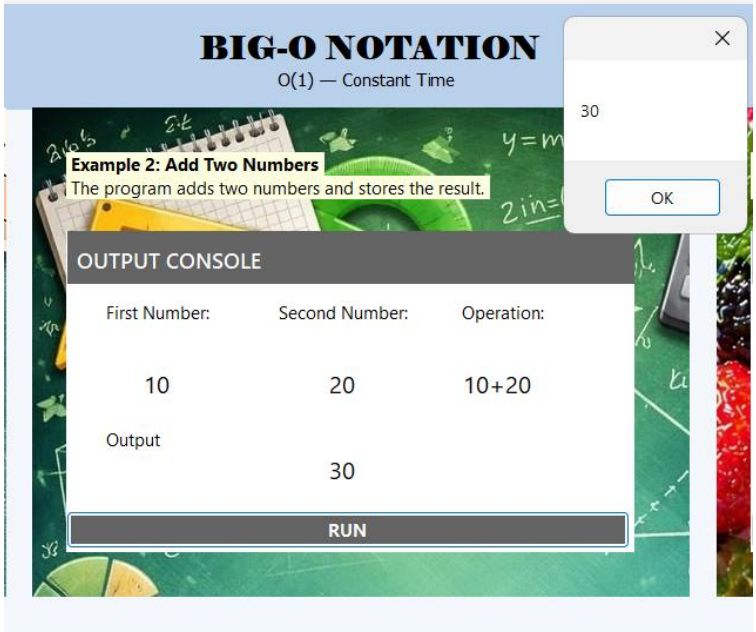


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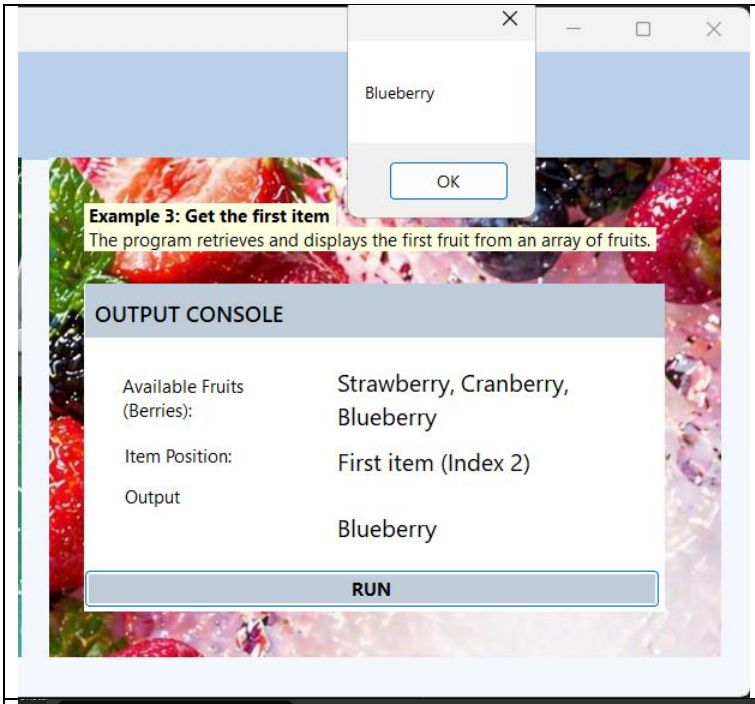
3. O(1) — Constant Time FINAL OUTPUT

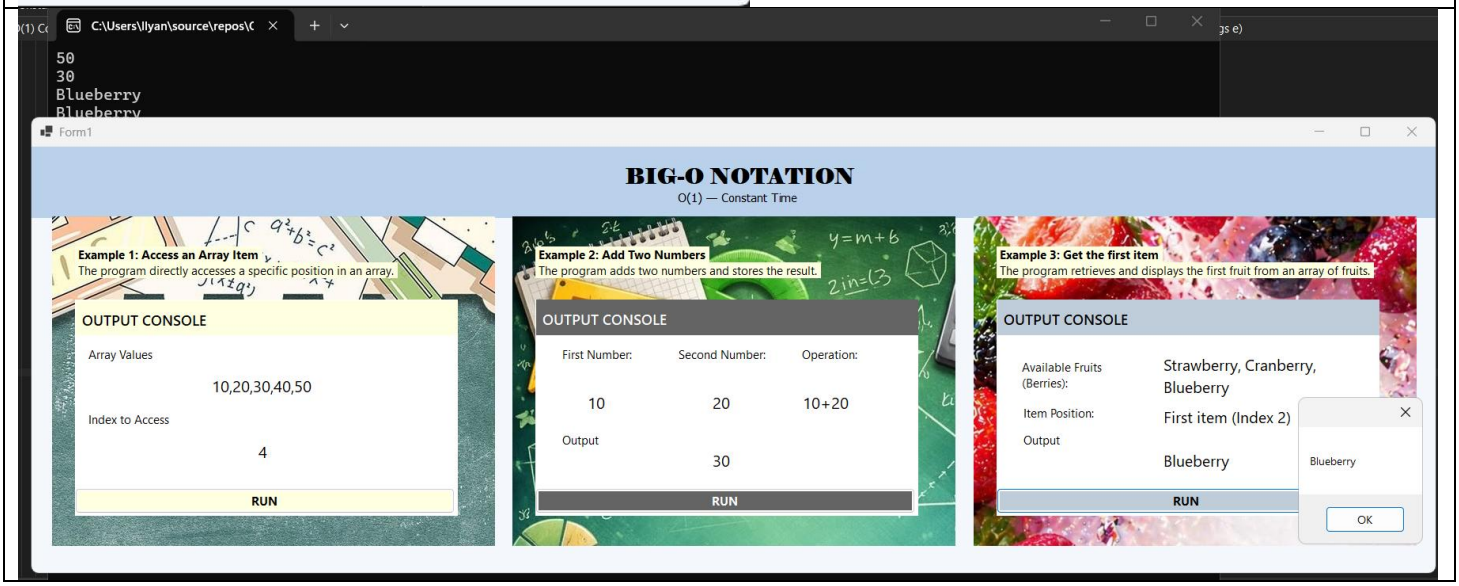




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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026

1. O(n) — Linear Time CODE

```
namespace O_n____Linear_Time
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnRun_Click(object sender, EventArgs e)
        {
            int[] numbers = { 10, 20, 30, 40, 50 };

            for (int i = 0; i < numbers.Length; i++)

            {
                Console.WriteLine(numbers[i]);
            }

        }

        private void button1_Click(object sender, EventArgs e)
        {
            string[] names = { "Ian", "Denzel", "Peter", "Lenette" }; // Professor's names; rearranged the order

            bool found = false; // Additional: Tracks whether the entered name is found

            foreach (string name in names) // Professor's original foreach loop
            {
                if (name == txtSearchName.Text) // Personalization: Uses the TextBox input instead of a fixed name like "Peter"
                {
                    found = true; // Additional: Marks the name as found
                    Console.WriteLine("Found!"); // Professor's original Console output
                    break; // Stops searching when the name is found
                }
            }

            if (found) // Additional: Checks if a match was found
            {
                lblResult.Text = "√ Found!"; // Additional: Displays the result in the Form
            }
            else
            {
                lblResult.Text = "X Name not found."; // Additional: Displays a message if the name is not found
            }
        }

```

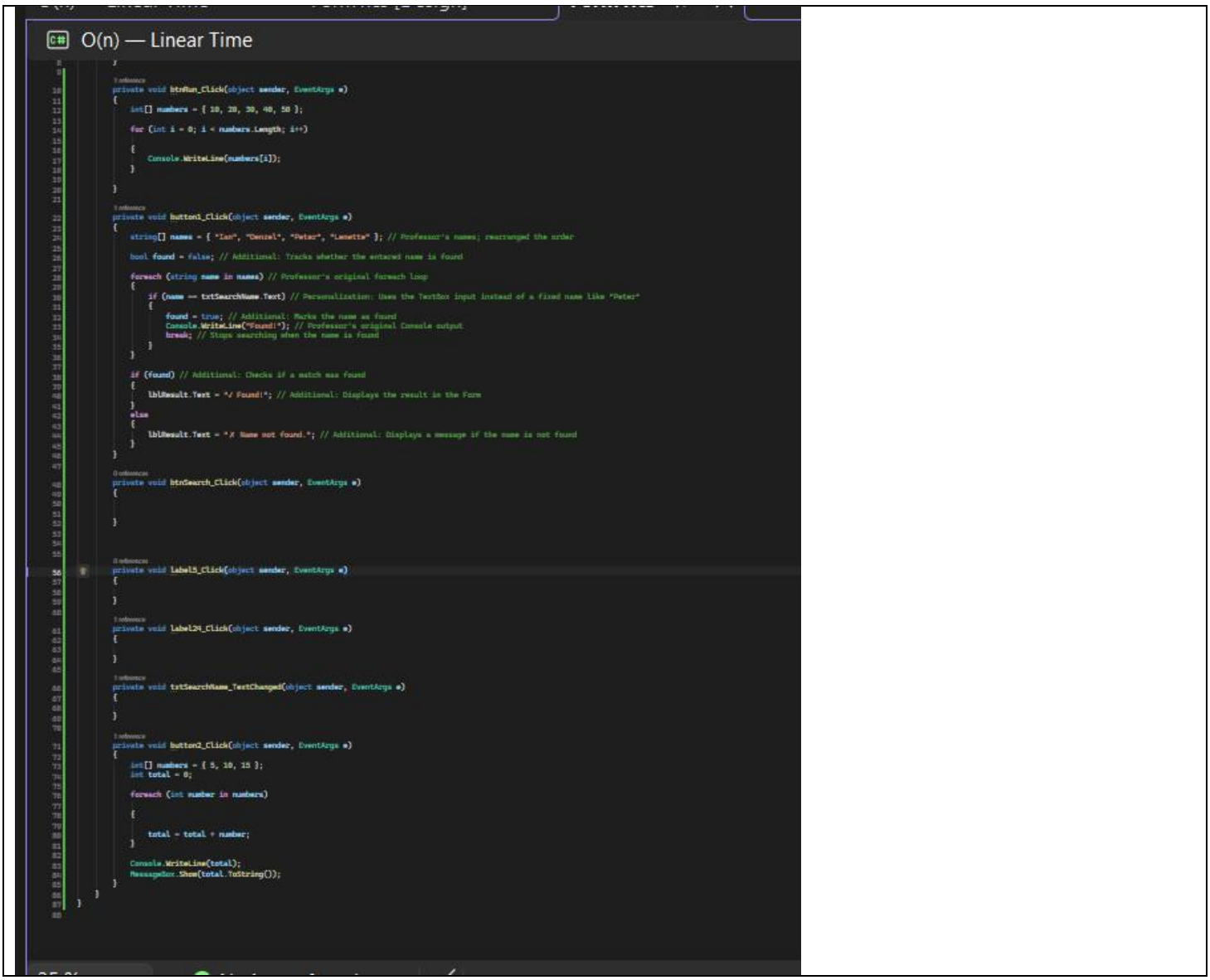
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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026

```
49.         private void btnSearch_Click(object sender, EventArgs e)
50.         {
51.
52.
53.         }
54.
55.
56.
57.         private void label5_Click(object sender, EventArgs e)
58.         {
59.
60.         }
61.
62.         private void label24_Click(object sender, EventArgs e)
63.         {
64.
65.         }
66.
67.         private void txtSearchName_TextChanged(object sender, EventArgs e)
68.         {
69.
70.         }
71.
72.         private void button2_Click(object sender, EventArgs e)
73.         {
74.             int[] numbers = { 5, 10, 15 };
75.             int total = 0;
76.
77.             foreach (int number in numbers)
78.
79.             {
80.
81.                 total = total + number;
82.             }
83.
84.             Console.WriteLine(total);
85.             MessageBox.Show(total.ToString());
86.         }
87.     }
88. }
```

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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026




```
        }

        1 reference
        private void btnRun_Click(object sender, EventArgs e)
        {
            int[] numbers = { 10, 20, 30, 40, 50 };

            for (int i = 0; i < numbers.Length; i++)

            {
                Console.WriteLine(numbers[i]);
            }

        }

        1 reference
        private void button1_Click(object sender, EventArgs e)
        {
            string[] names = { "Ian", "Denzel", "Peter", "Lenette" }; // Professor's names; rearranged the order

            bool found = false; // Additional: Tracks whether the entered name is found

            foreach (string name in names) // Professor's original foreach loop
            {
                if (name == txtSearchName.Text) // Personalization: Uses the TextBox input instead of a fixed name like "Peter"
                {
                    found = true; // Additional: Marks the name as found
                    Console.WriteLine("Found!"); // Professor's original Console output
                    break; // Stops searching when the name is found
                }
            }

            if (found) // Additional: Checks if a match was found
            {
                lblResult.Text = "✓ Found!"; // Additional: Displays the result in the Form
            }
            else
            {
                lblResult.Text = "✗ Name not found."; // Additional: Displays a message if the name is not found
            }
        }

        0 references
        private void btnSearch_Click(object sender, EventArgs e)
        {


        }



        0 references
        private void label5_Click(object sender, EventArgs e)
        {

        }

        1 reference
        private void label24_Click(object sender, EventArgs e)
        {

        }

        1 reference
        private void txtSearchName_TextChanged(object sender, EventArgs e)
        {

        }

        1 reference
        private void button2_Click(object sender, EventArgs e)
        {
            int[] numbers = { 5, 10, 15 };
            int total = 0;

            foreach (int number in numbers)

            {

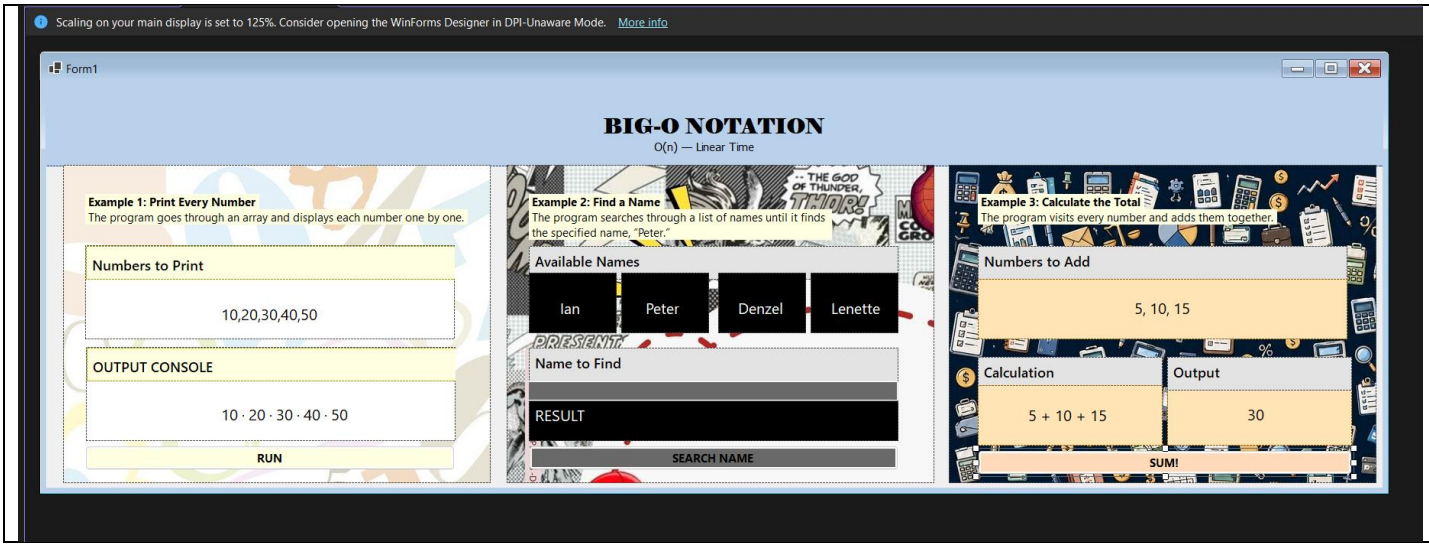
                total = total + number;
            }

            Console.WriteLine(total);
            MessageBox.Show(total.ToString());
        }
    }
}
```

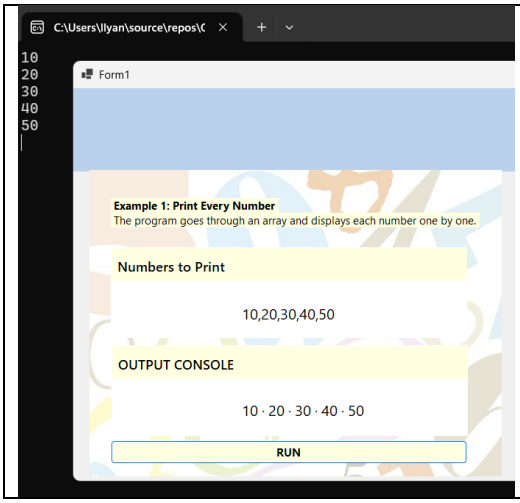
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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026

89. O(n) — Linear Time DESIGN

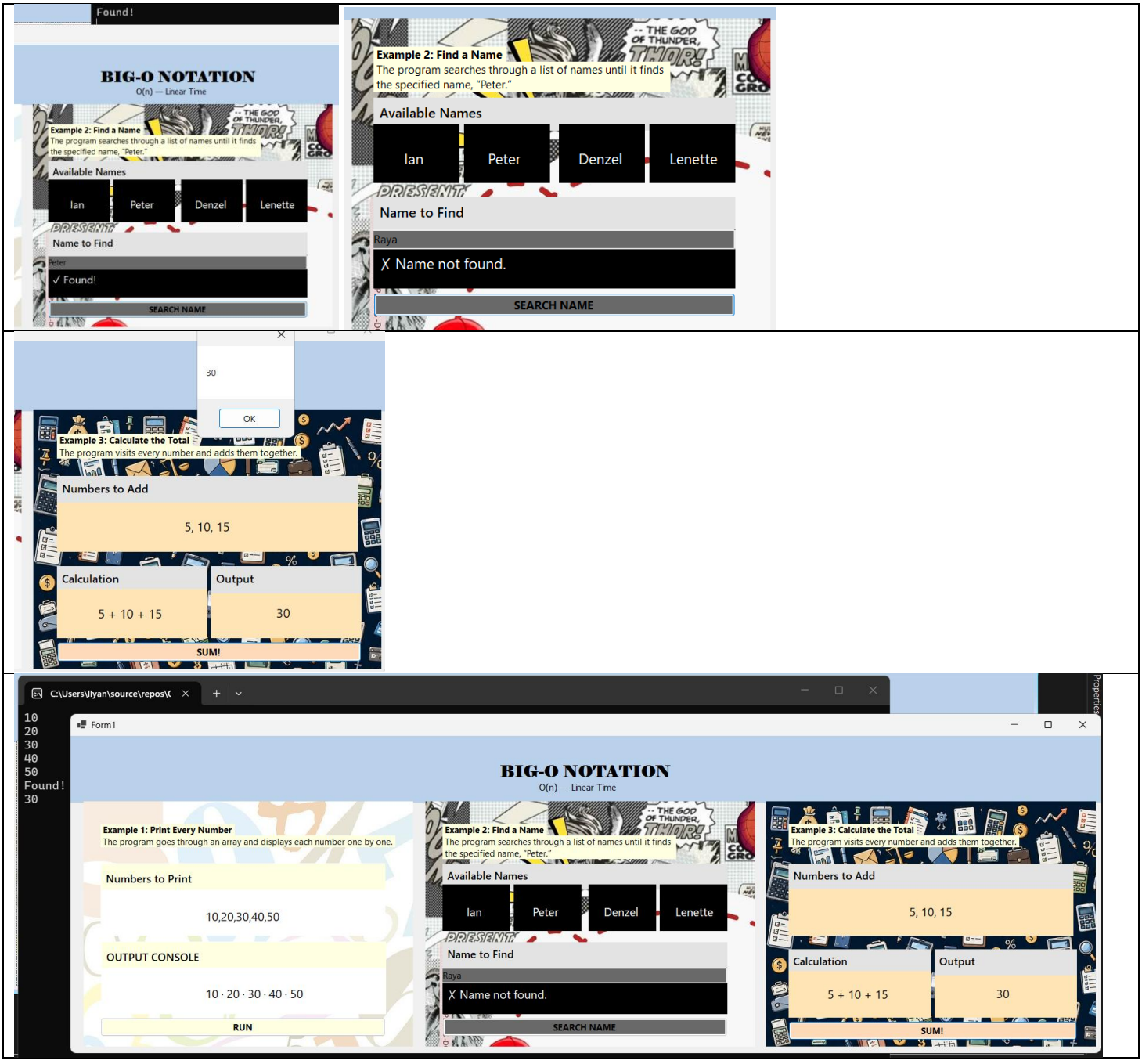


90. O(n) — Linear TimeFINAL OUTPUT



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DSTALGO
ANSWER SHEET
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1. O(log n) — Logarithmic Time CODE

```
2. using static System.Windows.Forms.VisualStyles.VisualStyleElement;
```

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```
3.
4.  namespace O_log_n____Logarithmic_Time
5.  {
6.      public partial class Form1 : Form
7.      {
8.          public Form1()
9.          {
10.             InitializeComponent();
11.         }
12.
13.         private void Form1_Load(object sender, EventArgs e)
14.         {
15.
16.         }
17.
18.         private void lblDescription_Click(object sender, EventArgs e)
19.         {
20.
21.         }
22.
23.         private void label14_Click(object sender, EventArgs e)
24.         {
25.
26.         }
27.
28.
29.         private void btnRun_Click(object sender, EventArgs e)
30.         // the additions let the user enter a number and see the result in the ListBox.
31.         {
32.             int[] numbers = { 10, 20, 30, 40, 50, 60, 70 };
33.
34.             int target = int.Parse(SEARCH.Text);
35.             int left = 0;
36.             int right = numbers.Length - 1;
37.             bool found = false;
38.
39.             Result.Items.Clear();
40.
41.             while (left <= right)
42.             {
43.                 int middle = (left + right) / 2;
44.
45.                 if (numbers[middle] == target)
46.                 {
47.                     Console.WriteLine("Found!");
48.                     Result.Items.Add("Found!");
49.                     found = true;
50.                     break;
51.                 }
52.                 else if (numbers[middle] < target)
53.                 {
54.                     left = middle + 1;
55.                 }
```

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ANSWER SHEET
Term 1 AY 2025 - 2026

```
56.                 else
57.                 {
58.                     right = middle - 1;
59.                 }
60.             }
61.
62.             if (!found)
63.             {
64.                 Console.WriteLine("Not found!");
65.                 Result.Items.Add("Not found!");
66.             }
67.         }
68.
69.         private void txtNumber_TextChanged(object sender, EventArgs e)
70.         {
71.
72.         }
73.
74.         private void button1_Click(object sender, EventArgs e)
75.         // Added TextBox and ListBox to allow user input and display output.
76.         {
77.             int n = int.Parse(txtNumber.Text);
78.
79.             ResultHere.Items.Clear();
80.
81.             while (n > 1)
82.             {
83.                 n = n / 2;
84.                 Console.WriteLine(n);
85.                 ResultHere.Items.Add(n);
86.             }
87.         }
88.
89.         private void textBox1_TextChanged(object sender, EventArgs e)
90.         {
91.
92.         }
93.
94.         private void label6_Click(object sender, EventArgs e)
95.         {
96.
97.         }
98.
99.         private void label10_Click(object sender, EventArgs e)
100.          {
101.
102.          }
103.
104.          private void button1_Click_1(object sender, EventArgs e)
105.          // Added a button to demonstrate binary search style guessing and display the result
    in a pop-up.
106.          {
107.              int low = 1;
```

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ANSWER SHEET
Term 1 AY 2025 - 2026

```
108.             int high = 100;
109.
110.             string result = "";
111.
112.             while (low <= high)
113.             {
114.                 int middle = (low + high) / 2;
115.
116.                 Console.WriteLine(middle);
117.                 result += middle + "\n";
118.
119.                 high = middle - 1;
120.             }
121.
122.             // Displays the pop-up result
123.             MessageBox.Show(
124.                 "Binary-Search-Style Guessing\n\n" +
125.                 "Guesses:\n" + result +
126.                 "\nBig-O: O(log n)",
127.                 "Guessing Result"
128.             );
129.         }
130.
131.         }
132.     }
```

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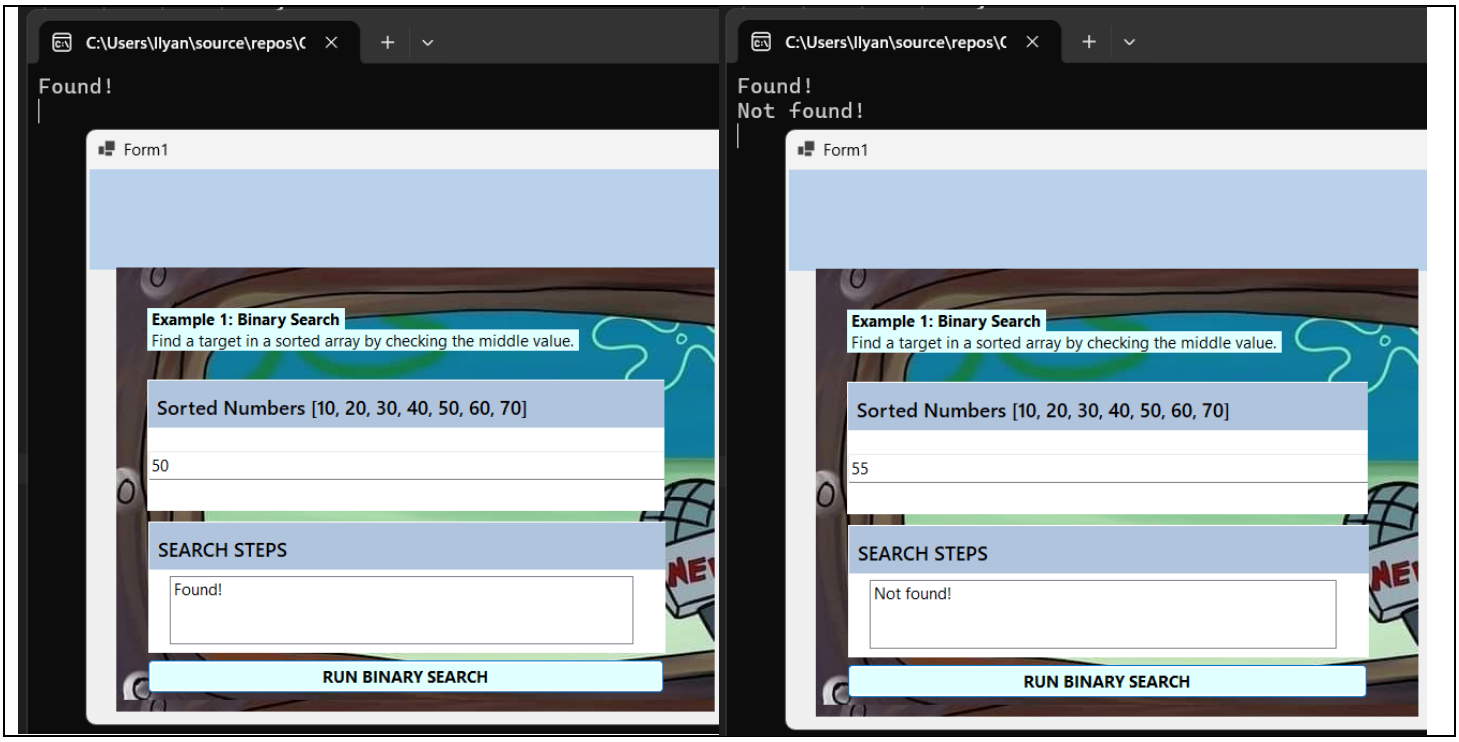
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ANSWER SHEET
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133. O(log n) — Logarithmic Time DESIGN

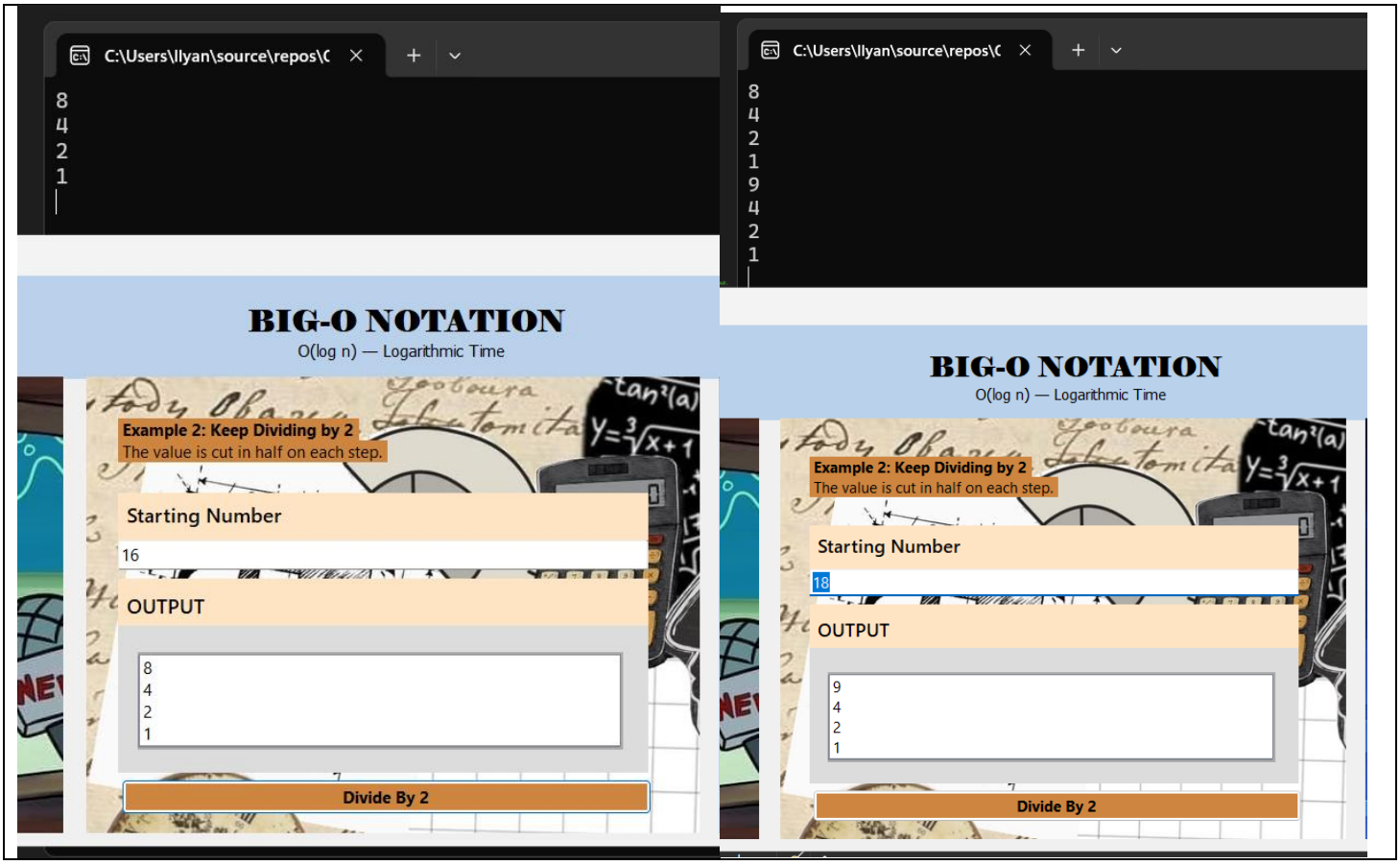


134. O(log n) — Logarithmic Time OUTPUT



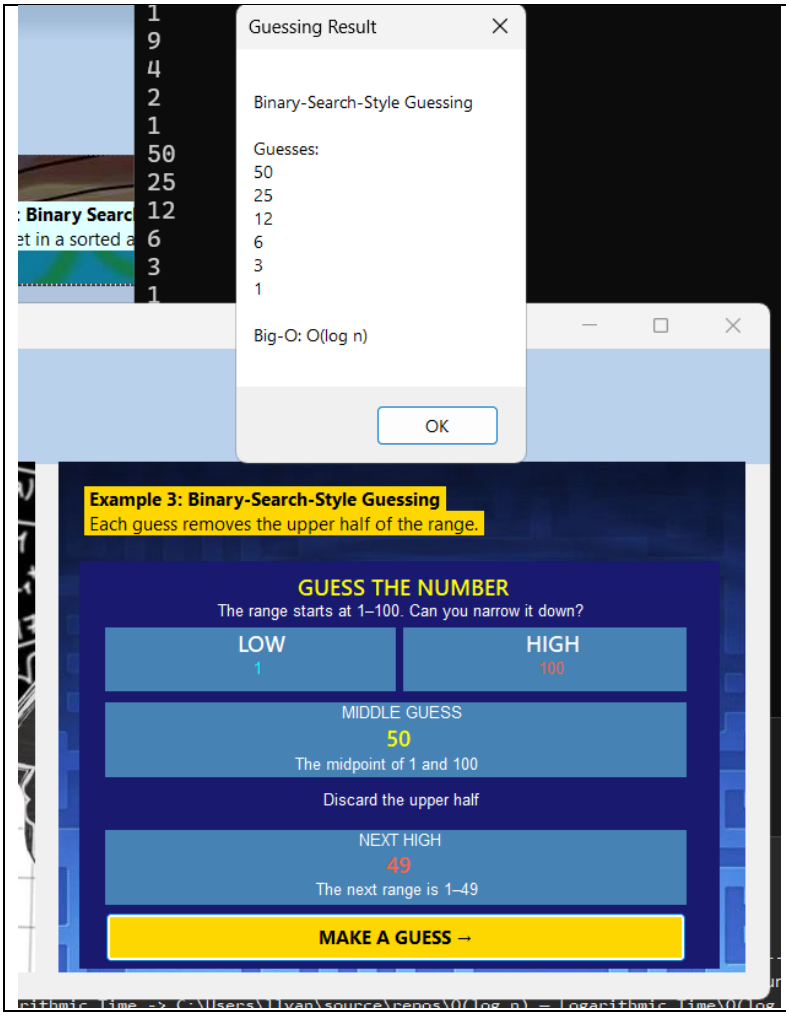
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Term 1 AY 2025 - 2026



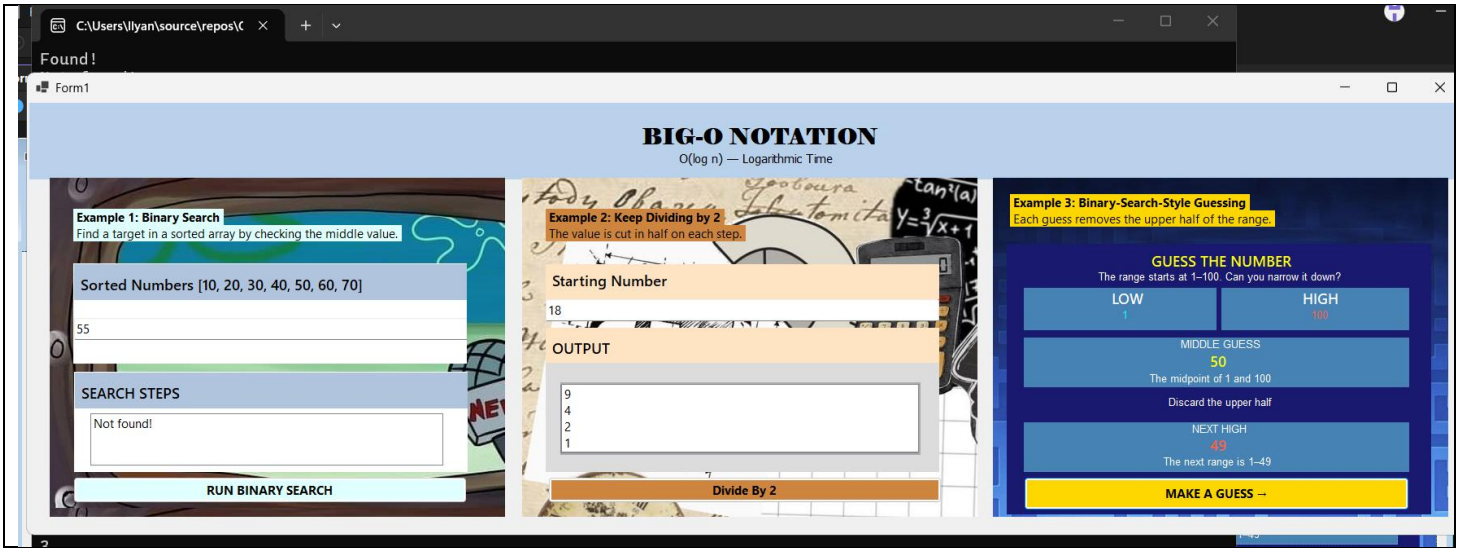
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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026



1. O(n²) — Quadratic Time CODE

```
2.  namespace O_n_____Quadratic_Time
3.  {
4.      public partial class Form1 : Form
5.      {
6.          public Form1()
7.          {
8.              InitializeComponent();
9.          }
10.
11.         private void label37_Click(object sender, EventArgs e)
12.         {
13.
14.         }
15.
16.         private void button3_Click(object sender, EventArgs e)
17.         {
18.             {
19.                 int[] numbers = { 1, 2, 3, 4, 5 };
20.
21.                 string output = ""; // Added
22.
23.                 for (int i = 0; i < numbers.Length; i++)
24.                 {
25.                     for (int j = 0; j < numbers.Length; j++)
26.                     {
27.                         Console.WriteLine(numbers[i] + " " + numbers[j]);
28.
29.                         output += numbers[i] + " " + numbers[j] + "\n"; // Added
30.                     }
31.                 }
```

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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026

```
32.
33.                 MessageBox.Show(output, "Compare Five Numbers"); // Added
34.             }
35.         }
36.
37.         private void panel7_Paint(object sender, PaintEventArgs e)
38.         {
39.
40.         }
41.
42.         private void button2_Click(object sender, EventArgs e)
43.         // Changed numbers = { 1, 2, 3 }; to colors = { "RED", "BLUE", "GREEN" };
44.         {
45.             string[] colors = { "RED", "BLUE", "GREEN" };
46.
47.             string output = "";
48.
49.             for (int i = 0; i < colors.Length; i++)
50.             {
51.                 for (int j = 0; j < colors.Length; j++)
52.                 {
53.                     Console.WriteLine(colors[i] + " vs " + colors[j]);
54.
55.                     output += colors[i] + " vs " + colors[j] + "\n";
56.                 }
57.             }
58.
59.             MessageBox.Show(output, "Color Matchup");
60.
61.         }
62.
63.
64.         private void label50_Click(object sender, EventArgs e)
65.         {
66.
67.         }
68.
69.         private void label30_Click(object sender, EventArgs e)
70.         {
71.
72.         }
73.
74.         private void label10_Click(object sender, EventArgs e)
75.         {
76.
77.         }
78.
79.         private void button1_Click(object sender, EventArgs e)
80.         // Changed name to Girls and values to { "Bubbles", "Blossom", "Buttercup" }
81.         {
82.             string[] Girls = { "Bubbles", "Blossom", "Buttercup" };
83.             string output = "";
84.
```

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DSTALGO
ANSWER SHEET
Term 1 AY 2025 - 2026

```
85.             for (int i = 0; i < Girls.Length; i++)
86.             {
87.                 for (int j = 0; j < Girls.Length; j++)
88.                 {
89.                     Console.WriteLine(Girls[i] + " checks " + Girls[j]);
90.                     output += Girls[i] + " vs " + Girls[j] + "\n";
91.                 }
92.             }
93.
94.             MessageBox.Show(output, "Girl Matchup");
95.         }
96.
97.
98.         private void Form1_Load(object sender, EventArgs e)
99.         {
100.
101.          }
102.
103.          private void button4_Click(object sender, EventArgs e)
104.          {
105.              string output = "";
106.
107.              for (int i = 1; i <= 5; i++)
108.              {
109.                  for (int j = 1; j <= 5; j++)
110.                  {
111.                      Console.WriteLine(i + " x " + j + " = " + (i * j));
112.                      output += i + " x " + j + " = " + (i * j) + "\n";
113.                  }
114.              }
115.
116.              MessageBox.Show(output, "Multiplication Table");
117.          }
118.      }
119. }
```

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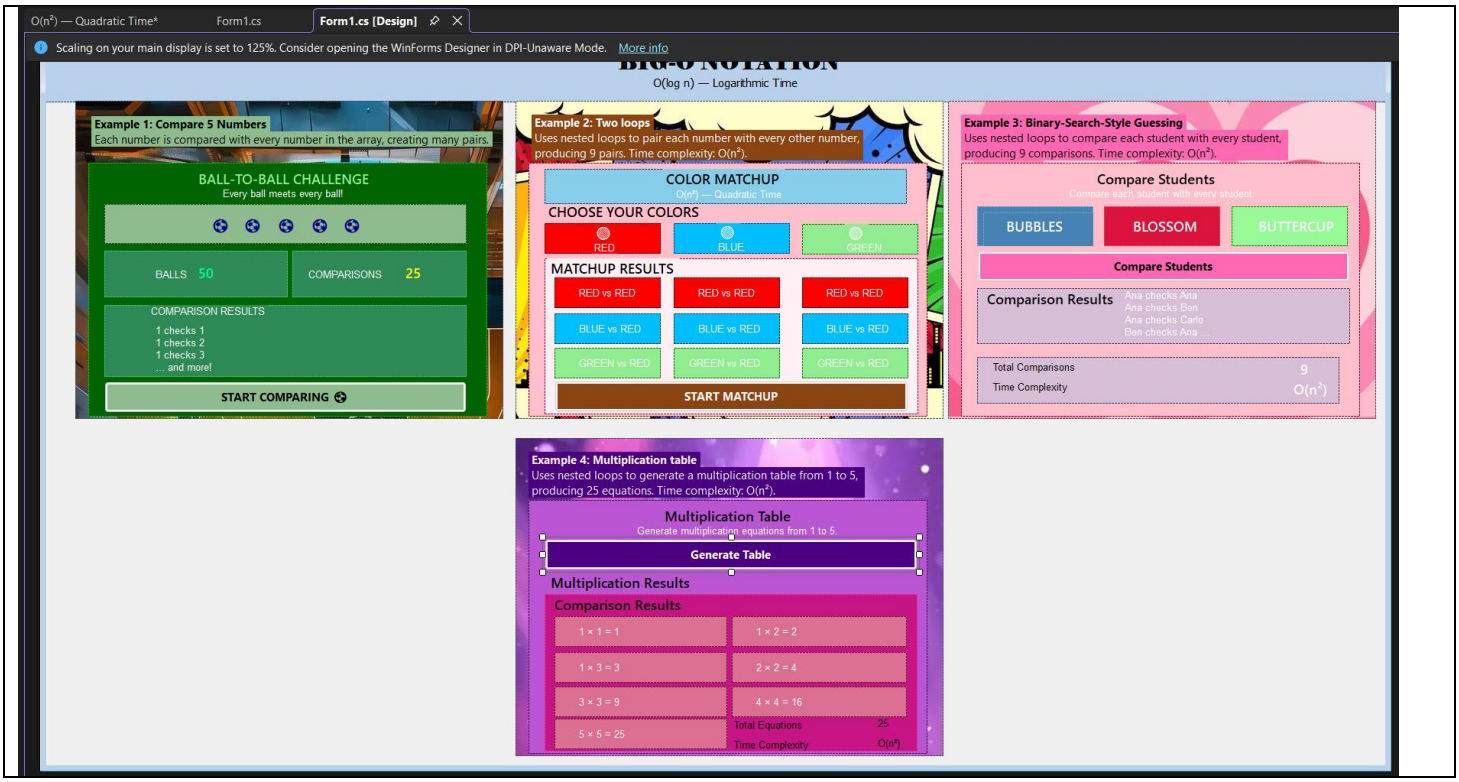
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Term 1 AY 2025 - 2026

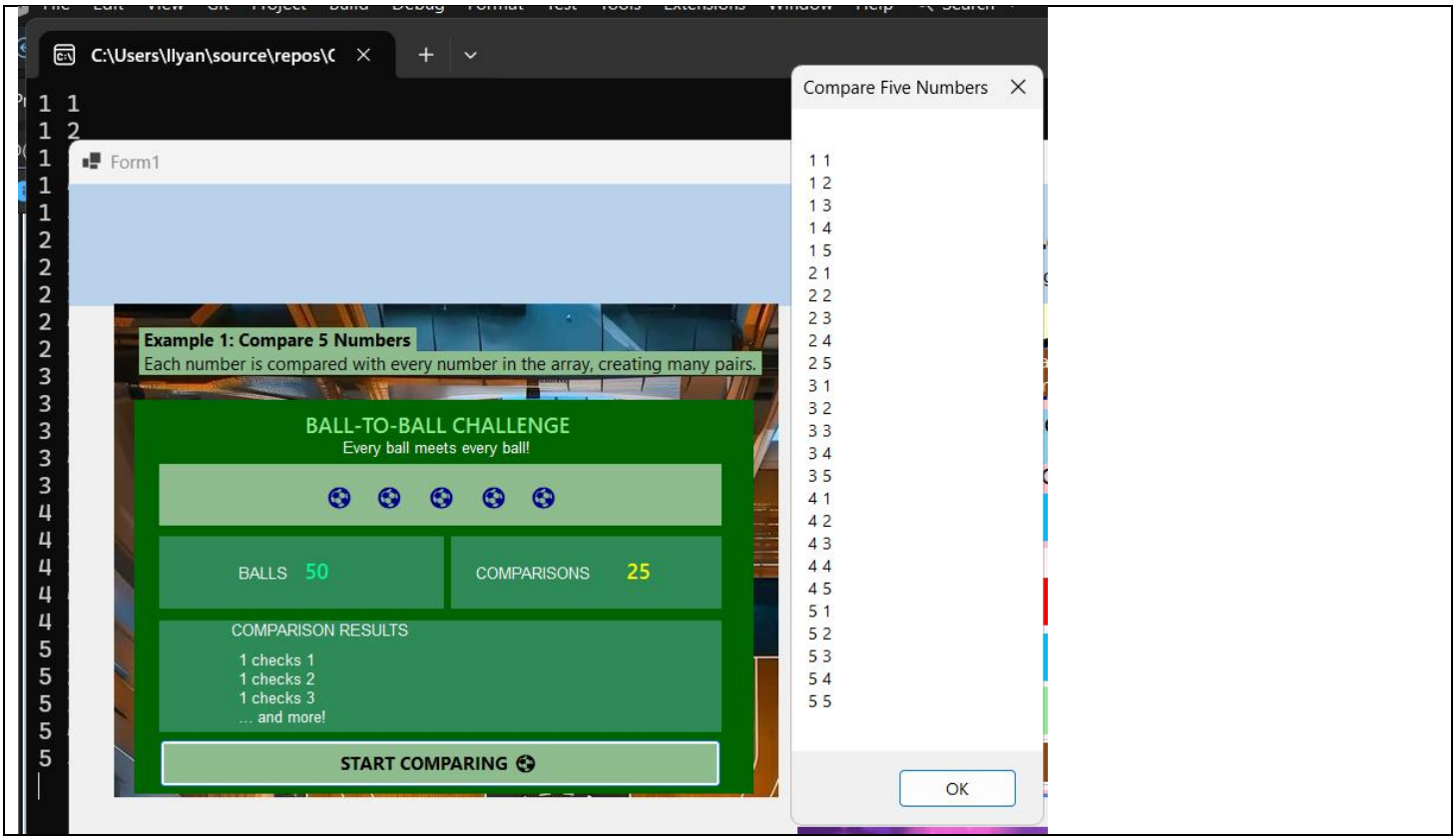
2. O(n²) — Quadratic Time DESIGN



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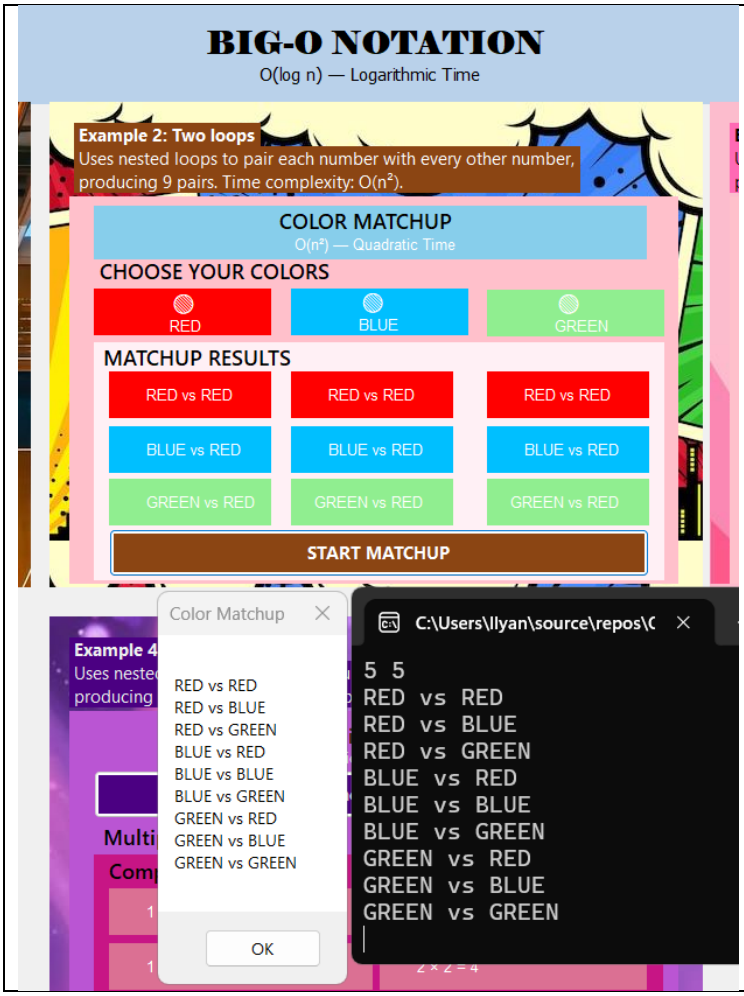
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3. O(n²) — Quadratic Time OUTPUT



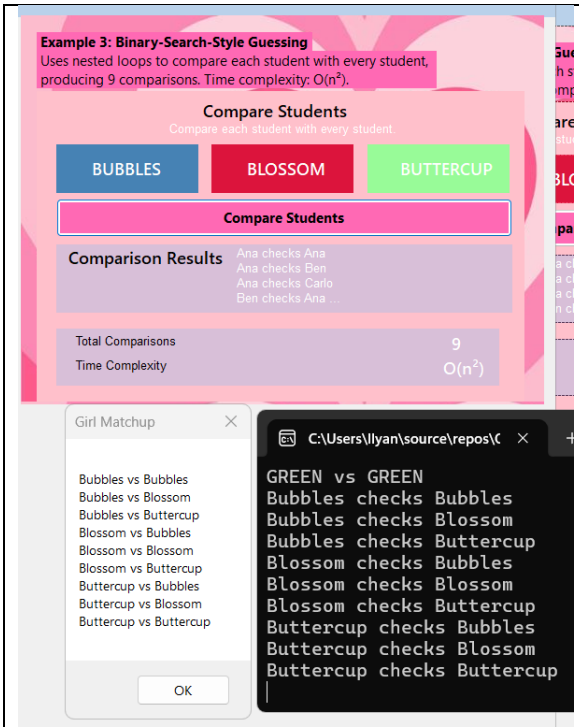
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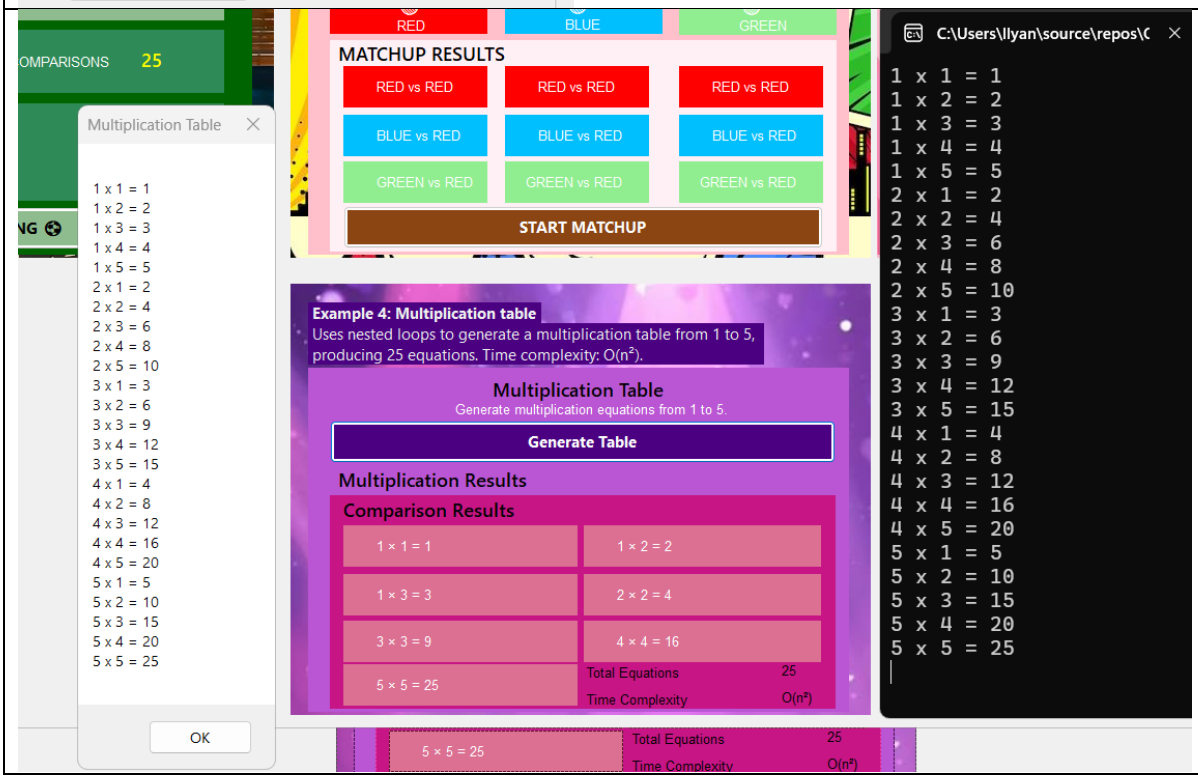
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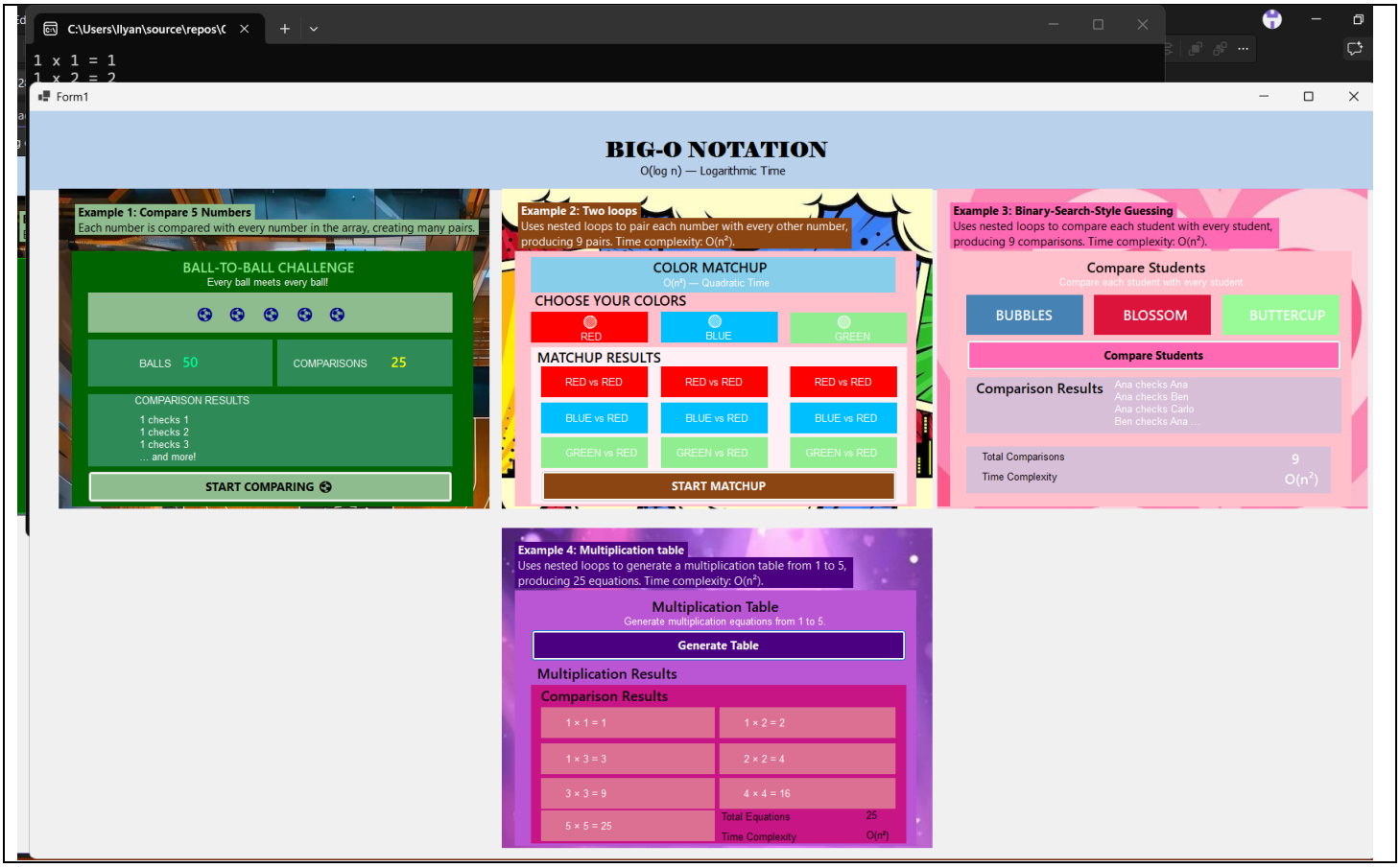
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Term 1 AY 2025 - 2026



1. O(2ⁿ) — Exponential Time CODE

```
2.  namespace O_n_____Quadratic_Time
3.  {
4.      public partial class Form1 : Form
5.      {
6.          public Form1()
7.          {
8.              InitializeComponent();
9.          }
10.
11.         private void label37_Click(object sender, EventArgs e)
12.         {
13.
14.         }
15.
16.         private void button3_Click(object sender, EventArgs e)
17.         {
18.             int n = 3;
```

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Term 1 AY 2025 - 2026

```
19.             int combinations = 1;
20.
21.             for (int i = 0; i < n; i++)
22.             {
23.                 combinations = combinations * 2;
24.             }
25.
26.             Console.WriteLine("Number of Combinations: " + combinations);
27.
28.             MessageBox.Show("Number of Combinations: " + combinations,
29.                             "Light Switch Challenge");
30.         }
31.
32.         private void panel7_Paint(object sender, PaintEventArgs e)
33.         {
34.
35.         }
36.
37.         private void button2_Click(object sender, EventArgs e)
38.         {
39.             int tosses = 5;
40.             int possibilities = 1;
41.
42.             for (int i = 0; i < tosses; i++)
43.             {
44.                 possibilities *= 2;
45.             }
46.
47.             Console.WriteLine("Possible results: " + possibilities);
48.
49.             MessageBox.Show("Possible results: " + possibilities,
50.                             "Coin Toss Challenge");
51.         }
52.
53.
54.         private void label50_Click(object sender, EventArgs e)
55.         {
56.
57.         }
58.
59.         private void label30_Click(object sender, EventArgs e)
60.         {
61.
62.         }
63.
64.         private void label10_Click(object sender, EventArgs e)
65.         {
66.
67.         }
68.
69.         private void button1_Click(object sender, EventArgs e)
70.         {
71.             int positions = int.Parse(txtpass.Text);
```

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ANSWER SHEET
Term 1 AY 2025 - 2026

```
72.             int possibilities = 1;
73.
74.             for (int i = 0; i < positions; i++)
75.             {
76.                 possibilities *= 2;
77.             }
78.
79.             Result.Items.Clear();
80.             Result.Items.Add("Possible Passwords: " + possibilities);
81.
82.             Console.WriteLine("Possible Passwords: " + possibilities);
83.
84.             MessageBox.Show("Possible Passwords: " + possibilities,
85.                             "Password Challenge");
86.         }
87.
88.
89.         private void Form1_Load(object sender, EventArgs e)
90.         {
91.
92.         }
93.
94.         private void button4_Click(object sender, EventArgs e)
95.         {
96.             string output = "";
97.
98.             for (int i = 1; i <= 5; i++)
99.             {
100.                for (int j = 1; j <= 5; j++)
101.                {
102.                    Console.WriteLine(i + " x " + j + " = " + (i * j));
103.                    output += i + " x " + j + " = " + (i * j) + "\n";
104.                }
105.            }
106.
107.            MessageBox.Show(output, "Multiplication Table");
108.        }
109.
110.        private void button4_Click_1(object sender, EventArgs e)
111.        {
112.            int n = int.Parse(txtCombo.Text);
113.
114.            COMBODATABASE.Items.Clear();
115.
116.            Choose(n, "");
117.
118.            MessageBox.Show("Total Combinations: " + Math.Pow(2, n),
119.                            "Combo Quest");
120.        }
121.
122.        void Choose(int n, string result)
123.        {
124.            if (n == 0)
```

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```
125.             {
126.                 Console.WriteLine(result);
127.                 COMBODATABASE.Items.Add(result);
128.                 return;
129.             }
130.
131.             Choose(n - 1, result + "A");
132.             Choose(n - 1, result + "B");
133.         }
134.     }
135. }
```

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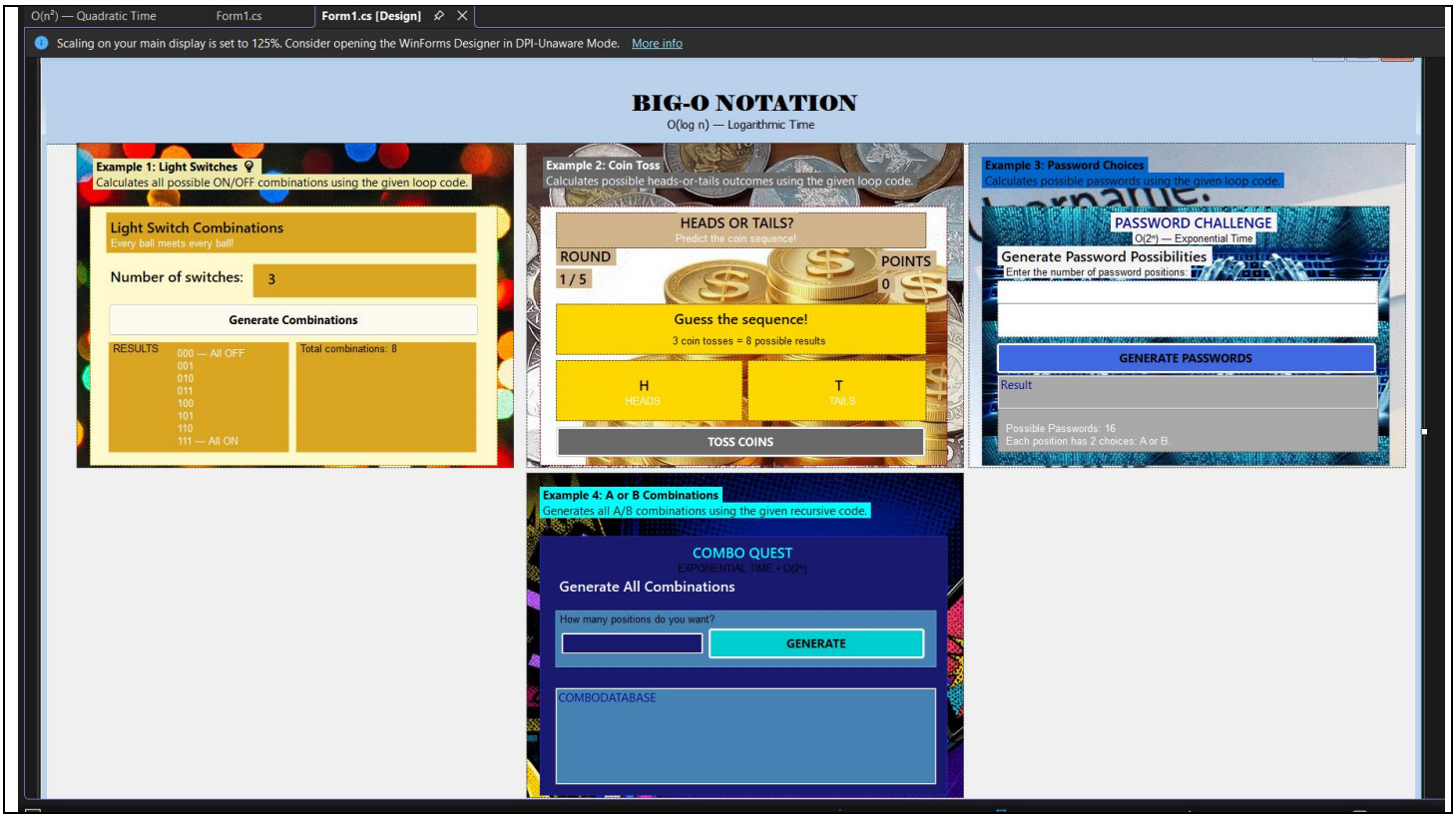
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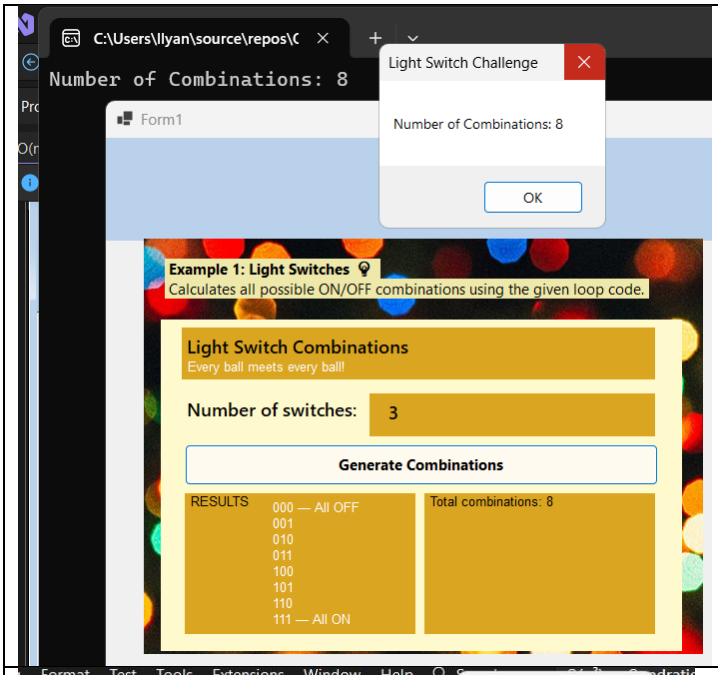
2. O(2ⁿ) — Exponential Time DESIGN

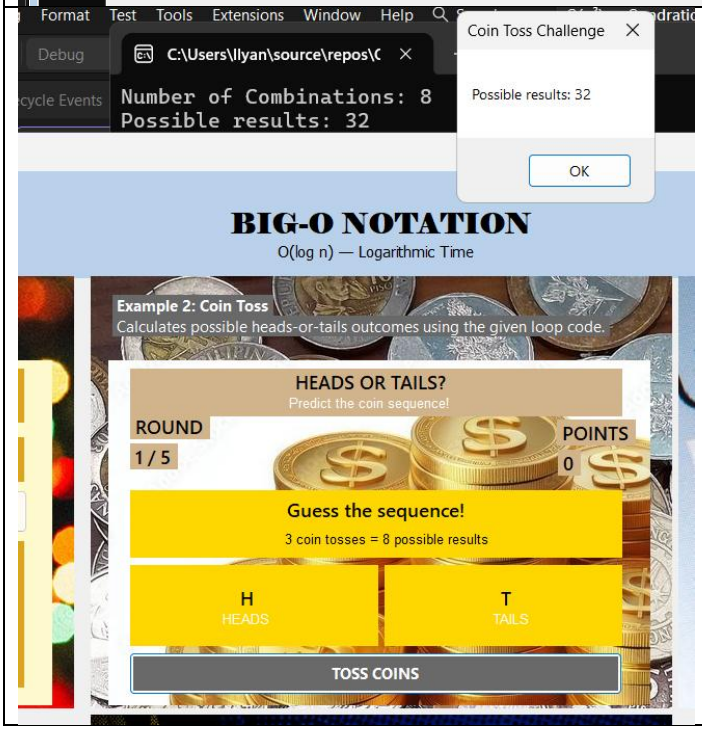


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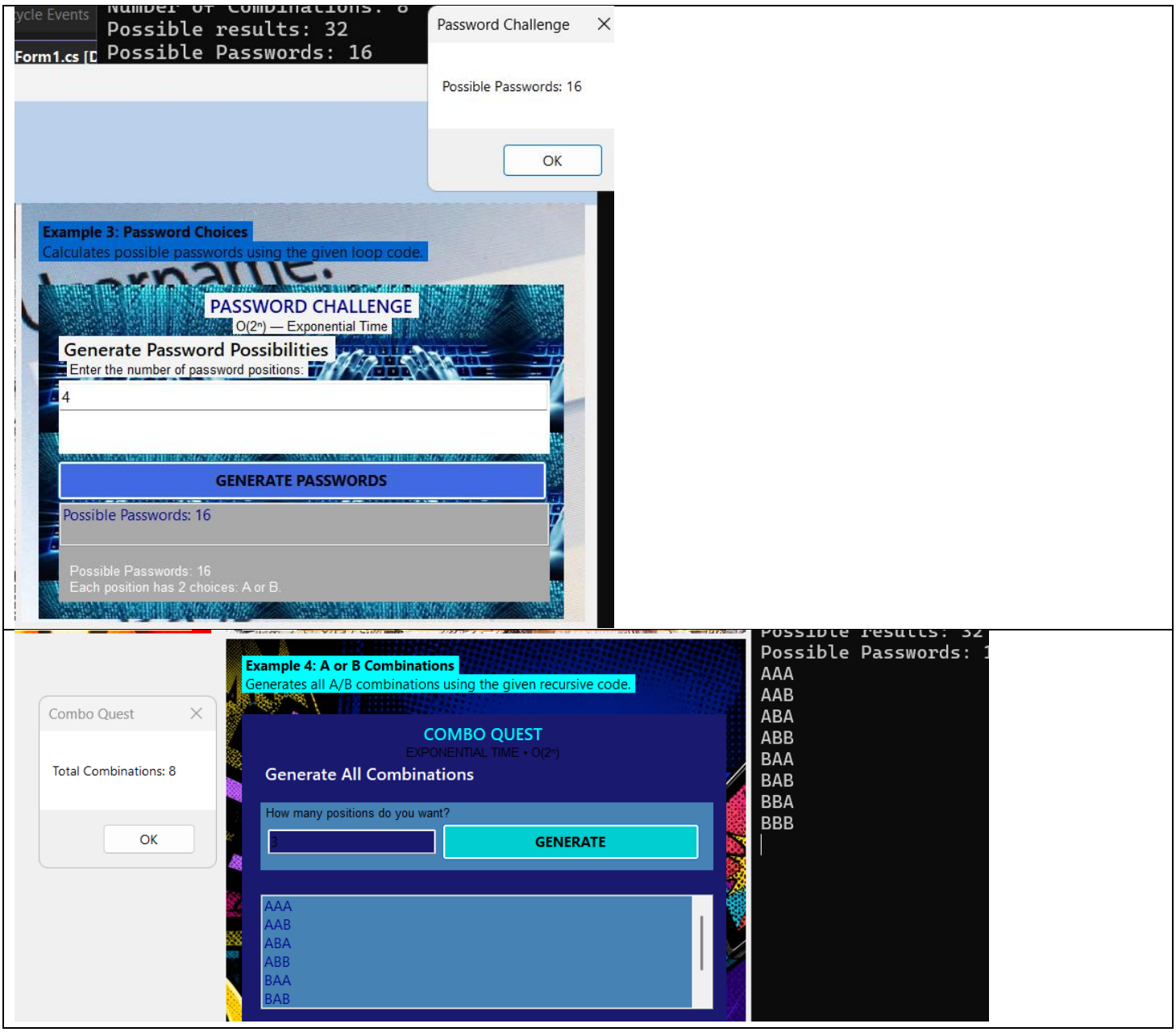
3. O(2ⁿ) — Exponential Time OUTPUT





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